

CO2-AT1-SCENARIOBASED QUESTIONS

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1.



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JAVA PROGRAMMING

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1924252018D

CO2-AT1-SCENARIOBASED QUESTIONS

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Q1
Student Registration System
10 marks

Q2
E-Commerce Cart System
10 marks

Q3
Library Management System
10 marks

Q4
Employee Data Processing
10 marks

Q5
Online Voting System
10 marks

4/5 answered

Q1 Student Registration System

10 marks

Suppose that a university system stores student data, where each student must be unique and can enroll in multiple courses.

(a) Enumerate three Collection interfaces suitable for this system.

(b) Design a Collection structure (diagram/representation) to store students and their enrolled courses.

(c) Write a Java program using HashSet and HashMap with Generics to:

- Add students
- Map students to courses
- Display data using Iterator

Your Code

```
1 import java.util.HashMap;
2 import java.util.HashSet;
3 import java.util.Iterator;
4 import java.util.Map;
5 import java.util.Set;
6 public class StudentRegistration {
7     public static void main(String[] args) {
8         Map<String, Set<String>> studentMap = new HashMap<>();
9         Set<String> courses1 = new HashSet<>();
10        courses1.add("Java Programming");
11        courses1.add("Database Systems");
12        studentMap.put("Student_101", courses1);
13        Set<String> courses2 = new HashSet<>();
14        courses2.add("Web Technology");
15        courses2.add("Java Programming");
16        studentMap.put("Student_102", courses2);
17        System.out.println("==== Student Registration Details ===");
18        Iterator<Map.Entry<String, Set<String>>> iterator = studentMap.entrySet().iterator();
19
20        while (iterator.hasNext()) {
21            Map.Entry<String, Set<String>> entry = iterator.next();
22            String studentId = entry.getKey();
23            Set<String> enrolledCourses = entry.getValue();
24
25            System.out.println("Student ID: " + studentId);
26            System.out.println("Enrolled Courses: " + enrolledCourses);
27            System.out.println("-----");
28        }
29    }
30 }
```

Input

stdin input

Output

```
==== Student Registration Details ===
Student ID: Student_101
Enrolled Courses: [Database Systems, Java Programming]
Student ID: Student_102
Enrolled Courses: [Web Technology, Java Programming]
```

2.



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5/5 answered

Q2 E-Commerce Cart System

10 marks

Suppose that an online shopping system maintains a cart where products are added, removed, and displayed in order.

(a) List three Collection classes applicable for cart management.

(b) Represent how cart items are stored using a List-based structure.

(c) Write a Java program using ArrayList and Iterator to:

- Add items
- Remove an item
- Display all items

Your Code

```
1 import java.util.*;
2
3 public class Main {
4     public static void main(String[] args) {
5
6         ArrayList<String> cart = new ArrayList<>();
7
8         cart.add("Laptop");
9         cart.add("Mouse");
10        cart.add("Keyboard");
11        cart.add("Headphones");
12
13        System.out.println("Items in Shopping Cart:");
14
15        Iterator<String> it = cart.iterator();
16
17        while (it.hasNext()) {
18            System.out.println(it.next());
19        }
20    }
21 }
```

Input

stdin input

Output

```
Items in Shopping Cart:
Laptop
Mouse
Keyboard
Headphones
```

Run

Save

3.

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4/5 answered

Q3 Library Management System

10 marks

Suppose that a library stores books with unique ISBN numbers and needs fast retrieval.

(a) Name three Map implementations in Java.

(b) Design a schema-like representation using Map for storing ISBN and book details.

(c) Write a Java program using HashMap to:

- Insert book details
- Retrieve a book
- Display all books

Your Code

```

1 import java.util.HashMap;
2 import java.util.Map;
3 public class Main {
4     static class Book {
5         String title;
6         String author;
7         Book(String title, String author) {
8             this.title = title;
9             this.author = author;
10        }
11        @Override
12        public String toString() {
13            return "Title: " + title + " by " + author;
14        }
15    }
16    public static void main(String[] args) {
17        Map<String, Book> library = new HashMap<>();
18        library.put("978-0134685991", new Book("Effective Java", "Joshua Bloch"));
19        library.put("978-0132035844", new Book("Clean Code", "Robert C. Martin"));
20        Book searchedBook = library.get("978-0134685991");
21        System.out.println("Retrieved Book: " + searchedBook + "\n");
22        System.out.println("All Books in Library ---");
23        for (Map.Entry<String, Book> entry : library.entrySet()) {
24            System.out.println("ISBN: " + entry.getKey() + " | Details: " + entry.getValue());
25        }
26    }
27 }
                
```

Run

Save

Input

stdin input

Output

```

Retrieved Book: "Effective Java"
by Joshua Bloch

--- All Books in Library ---
ISBN: 978-0134685991 | Details:
Effective Java by Joshua Bloch
                
```

4.

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4/5 answered

Q4 Employee Data Processing

10 marks

Suppose that a company processes employee records and filters employees based on salary.

(a) List three advantages of Generics in Collections.

(b) Represent how employee data is stored using List with Generics.

(c) Write a Java program using Iterator to:

- Store employee objects
- Display employees with salary > 50,000

Your Code

```

1 import java.util.ArrayList;
2 import java.util.Iterator;
3 import java.util.List;
4
5 public class Main {
6
7     static class Employee {
8         int id;
9         String name;
10        double salary;
11
12        Employee(int id, String name, double salary) {
13            this.id = id;
14            this.name = name;
15            this.salary = salary;
16        }
17
18        @Override
19        public String toString() {
20            return "ID: " + id + ", Name: " + name + ", Salary: $" + salary;
21        }
22    }
23
24    public static void main(String[] args) {
25        List<Employee> employeeList = new ArrayList<>();
26
27        employeeList.add(new Employee(101, "Alice", 62000));
28        employeeList.add(new Employee(102, "Bob", 45000));
29        employeeList.add(new Employee(103, "Charlie", 75000));
30        employeeList.add(new Employee(104, "David", 48000));
31    }
                
```

Run

Save

Input

stdin input

Output

```

--- Employees with Salary >
$50,000 ---
ID: 101, Name: Alice, Salary:
$62000.0
ID: 103, Name: Charlie, Salary:
$75000.0
                
```

5.



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Q5 Online Voting System
10 marks

Suppose that an online voting system ensures one vote per user and counts votes efficiently.

(a) Identify three suitable Collection types for this system.

(b) Design a structure using Set and Map for voters and vote counting.

(c) Write a Java program using HashSet and HashMap to:

- Record votes
- Prevent duplicate voting
- Display results

Your Code

```

1 import java.util.HashMap;
2 import java.util.HashSet;
3 import java.util.Map;
4 import java.util.Set;
5 public class Main {
6     private static Set<String> votedUsers = new HashSet<>();
7     private static Map<String, Integer> voteCountMap = new HashMap<>();
8     public static void castVote(String voterId, String candidate) {
9         if (votedUsers.contains(voterId)) {
10             System.out.println("Voting Denied: " + voterId + " has already voted.");
11             return;
12         }
13         votedUsers.add(voterId);
14         voteCountMap.put(candidate, voteCountMap.getOrDefault(candidate, 0) + 1);
15         System.out.println("Vote successfully cast by " + voterId + " for " + candidate);
16     }
17     public static void main(String[] args) {
18         castVote("Voter_01", "Alice");
19         castVote("Voter_02", "Bob");
20         castVote("Voter_02", "Alice");
21         castVote("Voter_01", "Bob");
22         System.out.println("\n--- Election Results ---");
23         for (Map.Entry<String, Integer> entry : voteCountMap.entrySet()) {
24             System.out.println("Candidate: " + entry.getKey() + " | Votes: " + entry.getValue());
25         }
26     }
27 }
          
```

Run
Save

Input

stdin input

Output

```

Vote successfully cast by
Voter_01 for Alice -
Vote successfully cast by
Voter_02 for Bob.
Vote successfully cast by
Voter_02 for Alice
Vote successfully cast by
Voter_01 for Bob
          
```